ThinkIT Solutions Pty

Retail Sales Analytics

SQL Data Analysis Case Study

Document Type:	Business Analytics Case Study
Prepared For:	Data Analyst — ThinkIT Solutions Pty
Requested By:	Executive Leadership Team
Dataset:	retail_sales_dataset.csv (1,000 transactions)
Period Covered:	01 January 2023 – 31 December 2023
Classification:	INTERNAL USE ONLY
Date Issued:	11 May 2026

Table of Contents

1.	Executive Brief & Business Context	3
2.	Dataset Overview	4
3.	Analytical Objectives	4
4.	Case Study Questions	5–11
4.1	Module A: Revenue & Sales Performance	5
4.2	Module B: Customer Demographics & Behaviour	7
4.3	Module C: Product & Category Insights	8
4.4	Module D: Operational & Temporal Trends	9
4.5	Module E: Strategic & Executive-Level Questions	10
5.	Deliverables & Presentation Requirements	11
6.	Evaluation Rubric	12
7.	Data Dictionary	12

1. Executive Brief & Business Context

ThinkIT Solutions Pty is a multi-category retail company operating across three core product verticals — **Beauty**, **Clothing**, and **Electronics**. With a growing customer base and an expanding transaction volume, the Executive Leadership Team has identified a critical need to leverage internal transactional data to drive better strategic decisions for the fiscal year ahead.

The Chief Executive Officer, in collaboration with the Chief Commercial Officer and the Head of Finance, has formally commissioned the Data Analytics Division to conduct a comprehensive review of the company's 2023 retail sales performance. The findings will be presented at the upcoming **Annual Strategic Review Board Meeting**.

Memo to the Analyst

You are hereby appointed as the Lead Data Analyst for this engagement. Your mandate is to query, analyse, and interpret the company's transactional sales data using SQL, and to produce a structured analytical report accompanied by clear visualisations and business-oriented commentary. You are expected to answer each business question listed in this document with a working SQL query, a result summary, and a written insight that leadership can act upon. All findings must be accurate, concise, and presented in plain business language. The board has zero tolerance for unsupported claims — every statement must be backed by data.

Important Notice: This case study is designed for a SQL analysis environment. All questions must be answered with executable SQL queries. Assume the dataset has been loaded into a relational database table named **retail_sales**. You may use standard SQL (compatible with PostgreSQL, MySQL, or SQLite). Marks will be awarded for query correctness, efficiency, and the quality of business commentary.

2. Dataset Overview

The dataset provided contains **1,000 retail transaction records** spanning the full calendar year 2023 (January to December). Each record captures a single customer purchase event.

Data Dictionary

Column Name	Data Type	Description
Transaction ID	INTEGER	Unique identifier for each sales transaction
Date	DATE	Date the transaction occurred (YYYY-MM-DD)
Customer ID	VARCHAR	Unique customer identifier (e.g. CUST001)
Gender	VARCHAR	Customer gender: Male or Female
Age	INTEGER	Customer age in years (range: 18–64)
Product Category	VARCHAR	Category sold: Beauty, Clothing, Electronics
Quantity	INTEGER	Units purchased per transaction (1–4)
Price per Unit	DECIMAL	Unit selling price in ZAR (R)

Total Amount	DECIMAL	Total value = Quantity × Price per Unit

3. Analytical Objectives

The Executive Leadership Team has structured this analytical engagement around five strategic areas of inquiry. Each objective maps directly to a module of questions within Section 4.

Objective	Business Goal	Module
O1 – Revenue Analysis	Identify revenue drivers, top periods, and overall financial health	**A**
O2 – Customer Insights	Understand who our customers are and how demographics influence spend	**B**
O3 – Product Performance	Determine which categories and price points perform best	**C**
O4 – Temporal Trends	Detect seasonality, peak trading months, and growth patterns	**D**
O5 – Strategic Intelligence	Synthesise insights for board-level decision making	**E**

4. Case Study Questions

The following 20 analytical questions have been formulated by the Executive Leadership Team. For each question, you are required to: **(1)** write a working SQL query, **(2)** present the query result in a formatted table or summary, and **(3)** provide a concise business insight (2–4 sentences) explaining what the data means and what action the business might consider.

Module A: Revenue & Sales Performance

The Finance Director needs a clear view of how revenue is being generated across the business.

1 What is the total revenue generated by the company in 2023?

Difficulty: Beginner

The CEO has opened the board presentation with a single question: 'How much money did we make last year?' Calculate the total revenue (Total Amount) across all 1,000 transactions for the full year 2023. Present this as a single summary figure and comment on whether this meets expectations for a retail business of this size.

SQL Concepts to Apply:

- *SUM() aggregate function*
- *Basic SELECT query*
- *Aliasing columns with AS*

2 What is the total revenue per product category?

Difficulty: Beginner

The Chief Commercial Officer wants to know which product categories are carrying the business. Break down total revenue by Beauty, Clothing, and Electronics. Rank the categories from highest to lowest revenue. Your insight should identify the strongest and weakest category and suggest whether the revenue distribution appears balanced or skewed.

SQL Concepts to Apply:

- *SUM() with GROUP BY*
- *ORDER BY DESC*
- *Percentage contribution (optional)*

3 What is the average transaction value, and how does it differ by category?

Difficulty: Beginner

The Head of Finance is concerned that some categories may have many small-ticket transactions that inflate volume but not profit. Calculate the average transaction value (Total Amount) overall and per product category. Identify which category commands the highest spend per transaction and what this means for margin strategy.

SQL Concepts to Apply:

- *AVG() aggregate function*
- *GROUP BY*
- *ROUND()*

4 Which are the top 10 highest-value transactions in the dataset?

Difficulty: Beginner

The Sales Director wants to understand what the highest-value individual purchases look like — who made them, what category they were in, and what quantity was purchased. Retrieve the top 10 transactions by Total Amount. Include Transaction ID, Customer ID, Date, Product Category, Quantity, and Total Amount. Are there any patterns in these high-value transactions?

SQL Concepts to Apply:

- *ORDER BY DESC*
- *LIMIT / TOP*
- *Multi-column SELECT*

5 What is the monthly revenue trend, and which month was the most profitable?

Difficulty: Intermediate

The Finance Director wants to present a month-by-month revenue trend chart at the board meeting. Aggregate total revenue by calendar month. Identify the peak revenue month and the lowest revenue month. Calculate the percentage difference between the two. Provide commentary on whether there is a seasonal pattern the business should plan for in 2024.

SQL Concepts to Apply:

- *EXTRACT() or strftime() or MONTH() for date parts*
- *GROUP BY month*
- *ORDER BY month*
- *Percentage difference calculation*

Module B: Customer Demographics & Behaviour

The Head of Marketing needs to understand who ThinkIT's customers are and what drives their purchasing decisions.

6 How does revenue and transaction count differ between male and female customers?

Difficulty: Beginner

The Marketing Director has commissioned a gender-based segmentation analysis. Compare male versus female customers across total revenue generated, total number of transactions, and average spend per transaction. Do female customers spend more per visit, or do they transact more frequently? Translate this into a recommendation for targeted marketing campaigns.

SQL Concepts to Apply:

- *GROUP BY Gender*
- *COUNT() and SUM()*
- *AVG()*
- *Multiple aggregates in one query*

7 Which age group contributes the most revenue to the business?

Difficulty: Intermediate

The Customer Experience team wants to identify which age demographic is most valuable to the business so that loyalty programmes and product assortment can be aligned accordingly. Segment customers into the following age bands: 18–25, 26–35, 36–45, 46–55, and 56–64. For each band, calculate total revenue, number of transactions, and average spend. Which age group should ThinkIT prioritise in its retention strategy?

SQL Concepts to Apply:

- *CASE WHEN ... THEN for age banding*
- *GROUP BY derived column*
- *SUM(), COUNT(), AVG()*

8 Which product category is most popular among female vs. male customers?

Difficulty: Intermediate

The Merchandising Manager has asked whether men and women shop differently across the three product categories. For each gender, determine the top product category by both number of transactions and total revenue. Present a comparison table and highlight any unexpected patterns in cross-gender category preferences.

SQL Concepts to Apply:

- *GROUP BY two columns*
- *COUNT() and SUM()*
- *Pivot-style analysis or ranked output*

9 What is the spending pattern of customers aged 18–25 (the youth segment)?

Difficulty: Intermediate

The Head of Marketing is building a Gen Z/Youth engagement strategy and needs to understand how young customers (aged 18 to 25) shop. Analyse this cohort's preferred product categories, average spend per transaction, total spend, and total number of transactions. Compare their average spend to the overall company average and comment on the opportunity or risk this segment represents.

SQL Concepts to Apply:

- *WHERE Age BETWEEN*
- *GROUP BY*
- *Subquery or CTE for comparison value*

10 Who are the top 5 highest-spending customers overall?

Difficulty: Intermediate

The Loyalty Programme Manager wants to identify the company's most valuable customers for inclusion in a premium rewards tier. Identify the top 5 customers by total spend. Include their Customer ID, total spend, total number of transactions, average spend per visit, and predominant product category. What does their profile suggest about the company's most loyal customer archetype?

SQL Concepts to Apply:

- *GROUP BY Customer ID*
- *SUM(), COUNT(), AVG()*
- *ORDER BY total spend DESC*
- *LIMIT 5*

Module C: Product & Category Insights

The Chief Commercial Officer requires granular visibility into product performance across all three categories.

11 What are the most frequently purchased quantities, and do they vary by category?

Difficulty: Beginner

The Operations Manager is reviewing stock replenishment cycles and wants to know how customers typically buy. Determine the distribution of purchase quantities (1, 2, 3, 4 units) across all transactions, and then break this down by product category. Which quantity size is most common overall and within each category? How should this influence stock-holding strategy?

SQL Concepts to Apply:

- *GROUP BY Quantity*
- *COUNT()*
- *Nested GROUP BY with two columns*

12 What is the total quantity sold per product category, and which moves the most units?

Difficulty: Beginner

The Supply Chain Director wants to align procurement planning with actual sales velocity. Calculate the total units sold per product category. Distinguish between unit volume and revenue contribution — a category may sell many units at low prices, or fewer units at high prices. What does this mean for purchasing decisions and warehouse allocation?

SQL Concepts to Apply:

- *SUM(Quantity) per category*
- *GROUP BY*
- *Comparison of volume vs. revenue*

13 What is the revenue contribution percentage of each product category?

Difficulty: Intermediate

The CFO wants to present a clean revenue breakdown in percentage terms at the board meeting. For each product category, calculate its total revenue AND its percentage share of total company revenue. Format the output cleanly. Identify whether the business is over-reliant on any single category, and what this means for business risk if that category underperforms.

SQL Concepts to Apply:

- *SUM() with subquery or CTE for total*
- *Percentage calculation: (category_revenue / total_revenue) * 100*
- *ROUND()*

14 What are the most and least profitable price points within each category?

Difficulty: Advanced

The Pricing Strategist has asked a nuanced question: within each product category, which price points (Price per Unit) generate the highest and lowest total revenue? List all unique price points per category with their associated total revenue and transaction count. Which price points should be promoted more aggressively, and which may need to be reconsidered?

SQL Concepts to Apply:

- *GROUP BY category + price per unit*
- *SUM() and COUNT()*
- *ORDER BY within groups using subquery or window function*

Module D: Operational & Temporal Trends

The Operations and Strategy teams need to understand the time-based dynamics of the business to plan staffing, promotions, and inventory cycles effectively.

15 Which quarter of 2023 recorded the highest revenue and transaction volume?

Difficulty: Intermediate

The CFO wants to identify which financial quarter was the strongest for ThinkIT Solutions. Assign each transaction to Q1 (Jan–Mar), Q2 (Apr–Jun), Q3 (Jul–Sep), or Q4 (Oct–Dec). For each quarter, calculate total revenue and number of transactions. Is the business back-loaded (stronger in Q4) or evenly spread? What operational planning implications does this create?

SQL Concepts to Apply:

- *CASE WHEN with MONTH() for quarter assignment*
- *GROUP BY quarter*
- *SUM() and COUNT()*

16 Which month had the highest number of transactions (not revenue)?

Difficulty: Beginner

The Operations Manager is building the 2024 staffing schedule and needs to know when the store is busiest by transaction volume, irrespective of the value of those transactions. Count the number of transactions per month. Present results ordered by month number. Is the busiest month by transaction count the same as the busiest month by revenue? If not, what does the discrepancy reveal?

SQL Concepts to Apply:

- *COUNT() per month*
- *EXTRACT() / MONTH() / strftime()*
- *GROUP BY and ORDER BY*

17 Is there a day-of-week pattern in purchasing behaviour?

Difficulty: Intermediate

The Marketing Manager wants to know whether certain days of the week see more sales activity, which could inform the scheduling of promotional campaigns and email marketing. Extract the day of the week from each transaction date and calculate total transactions and total revenue per day of the week. Present from Monday to Sunday. Are weekends significantly different from weekdays?

SQL Concepts to Apply:

- *DAYOFWEEK() / strftime('%w') / EXTRACT(DOW)*
- *GROUP BY day*
- *Ordering by weekday number*

18 What is the running cumulative revenue across the months of 2023?

Difficulty: Advanced

The Finance Director wants to track year-to-date (YTD) revenue accumulation — a common metric used in executive dashboards to show progress against annual targets. Calculate monthly revenue and then the running cumulative total from January through December. At which month did the company cross the 50% mark of annual revenue? Present this as a table with both monthly and cumulative figures.

SQL Concepts to Apply:

- *SUM() OVER (ORDER BY month) — window function*
- *Running total / cumulative sum*
- *CTE or subquery approach*

Module E: Strategic & Executive-Level Questions

These questions require multi-step SQL thinking and synthesis. They are designed to test the analyst's ability to connect data findings to real business decisions.

19 Which customer segments (gender x age group) generate the highest revenue per transaction?

Difficulty: Advanced

The CEO wants to identify the single most valuable customer persona in the business — the combination of gender and age group that spends the most per transaction on average. Create a cross-segment analysis combining Gender and Age Group (use the same 5 bands as Q7). For each combination, show average spend per transaction, total revenue, and transaction count. Rank segments from highest to lowest average spend. Who is ThinkIT's highest-value customer profile, and how should the business serve them better?

SQL Concepts to Apply:

- *CASE WHEN for age banding*
- *GROUP BY two dimensions*
- *AVG(), SUM(), COUNT()*
- *ORDER BY AVG DESC*

20 Build an executive summary table: one row per product category showing all key metrics.

Difficulty: Advanced

The CEO has requested a single-slide executive summary that can be presented to the board without any scrolling. For each product category, produce a single summary row containing: total revenue, total transactions, total units sold, average revenue per transaction, average units per transaction, top customer gender (by transaction count), and revenue share (%). This query will require combining multiple aggregations in one statement. Your insight should be a 4–6 sentence board-ready paragraph summarising ThinkIT's 2023 retail performance.

SQL Concepts to Apply:

- *Multiple aggregates in one SELECT*
- *Subqueries or CTEs*
- *CASE or subquery for mode (top gender)*
- *Percentage calculation*

5. Deliverables & Presentation Requirements

Upon completion of the analytical work, the Lead Data Analyst is required to submit the following deliverables to the Executive Leadership Team. All deliverables are due by the date specified in the project brief issued by the Analytics Division Director.

D1	**SQL Script File (.sql)**
	A clean, commented SQL script containing all 20 queries. Each query must be preceded by a comment block identifying the question number, title, and a one-line description of the approach. Queries must be executable without modification on the provided dataset.
D2	**Analytical Report (PDF or .md)**
	A structured report containing: an executive summary, methodology notes, all query results presented as formatted tables, and written business insights for each question. The report must be professional in tone and free of technical jargon where possible.
D3	**Data Visualisations**
	At least 5 charts or visual representations of key findings (e.g. monthly revenue trend, category revenue split, gender comparison, age segment heatmap). Visuals may be created in any BI tool (Power BI, Tableau, Excel, Python/Matplotlib, etc.) and must be embedded in the analytical report.
D4	**Executive Presentation Deck (optional but recommended)**
	A slide deck (5–10 slides) summarising the most critical findings for board consumption. This should include only the most impactful insights, formatted for a non-technical audience. One slide must address a strategic recommendation based on the data.
D5	**GitHub Repository (public)**
	All work must be uploaded to a public GitHub repository with a professional README.md file that describes the project context, dataset, tools used, and instructions for running the SQL scripts. The repository structure should demonstrate good project organisation and version control practices.

6. Evaluation Rubric

Each question will be evaluated according to the following criteria. The total project is scored out of 100 marks.

Criterion	Description	Weight
SQL Correctness	The query runs without errors and returns the expected output	35%
Query Efficiency	The query avoids redundancy; uses appropriate clauses and functions	15%
Result Presentation	Output is clearly displayed in a readable, formatted table	15%
Business Insight	Written commentary is relevant, accurate, and actionable	25%
Code Readability	Queries are commented, formatted, and use meaningful aliases	10%

Difficulty Weighting

Beginner questions (Q1–Q4, Q6, Q11, Q12, Q16) each carry equal base weight. Intermediate questions carry 1.2x the base mark. Advanced questions (Q14, Q18, Q19, Q20) carry 1.5x the base mark. The total mark is normalised to 100 at the end of grading.

Good luck! The Executive Leadership Team at ThinkIT Solutions Pty looks forward to reviewing your findings. Remember: great data analysis is not just about writing correct SQL — it is about telling the business a story it can act on. We expect you to be that storyteller.